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MakeUpMirror: mirroring make-ups and verifying faces post make-up

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Abstract: Facial make-up changes the appearance of a person and significantly degrades the performance of automated face verification (FV) systems. Here, the authors propose the design of an end-to-end siamese convolutional neural network (SCNN) that simultaneously replicates the facial make-up of a subject using its target image (under facial make-up) on a query face image and verifies the identity of the query face sample either with or without make-up. The SCNN model is designed using loss functions to deal with the variations due to make-up. The proposed architecture can reciprocate the make-up at appropriate locations of the face without any human interventions. Rigorous experimentations on four benchmark facial make-up datasets reveal the efficiency of their proposed model. Ablation studies show improvement of 4% for genuine acceptance rate at 0.1% false acceptance rate and reduction of equal error rate by 42% for FV in case of YouTube Make-up dataset, and '10%' in case of Virtual Make-up dataset, when compared to the nearest state-of-the-art method. For the transfer of make-up, the similarity measures also show the effectiveness of their method, where the peak signal-to-noise ratio and structural similarity values show an improvement by ∼20–24 and ∼29–32%, respectively, when compared to a recent state-of-the-art technique.

1 Introduction

Automatic face recognition (FR) finds applications in several fields such as surveillance, biometric authentication, person re-identification, image tagging, and human computer interaction [1]. The success of automated FR has improved steadily in the past decade, since the development of robust feature representation and classification techniques [2–5], as evident by increase in the performance on several publicly available datasets (e.g. PIE [6], LFW [7], FERET [8], YTF [9] etc.). However, heterogeneous FR still poses several challenges to the researchers since it involves cross-domain matching of fundamentally different images, e.g. camouflage, visible versus infra-red image matching, sketches versus photographs, or images captured by two different cameras.

A similar scenario arises in the problem of matching faces altered by the application of facial cosmetics (i.e. make-up). The images are captured under unconstrained environments, referred to as make-up in the wild, where make-up can severely degrade the performance of any automatic FR system. The problem of face verification (FV, as a special case of FR) using faces before and after make-up concerns the problem of cross-domain face-matching. Synthesising realistic make-ups on a face image require advanced editing skills beyond the abilities of casual photographers. This observation is one motivation for our work, where a technique is proposed in this paper to transfer the cosmetic appearance as on the face of the target (subject/individual) onto another face given as query image, without altering the subject ID of the query.

Cross-domain matching of data has produced a surge of transfer-learning (TL) techniques [10–14] in the field of machine learning. Similarly, domain adaptation (DA), as a special case of TL, has played a vital role in the field of heterogeneous FV, where the pair of faces to be matched varies in facial features. DA attempts to reduce the gap between the feature distributions captured from these faces with different appearances. Deep learning (DL) techniques [15], on the other hand, have caught the attention of the vision researchers since they produce appreciable improvement of accuracy in the field of FV [16].

In this paper, we propose an end-to-end siamese convolutional neural network (SCNN) that simultaneously solves two problems as follows: (i) verification of face images of an individual with or without make-up and (ii) transfer of make-up from one individual to another in an unconstrained environment. The work presented in this paper has three major contributions: (i) a novel SCNN architecture to solve both these problems simultaneously (i.e. the network not only applies make-up on a query face, but also verifies a query face without make-up with respect to the target face with make-up); (ii) a cross-dataset make-up application, such that the cosmetic changes can be applied to any face; and (iii) rigorous experimentations on three benchmark make-up datasets, viz. YMU (YouTube Make-up) [17], VMU (Virtual Make-up) [17, 18], and MIW (Make-up in wild) [17] to show the effectiveness of our proposed technique for make-up transfer and FV.

Next, Section 2 gives a literature review of the relevant works proposed in the recent past. Section 3 describes the proposed SCNN architecture used to solve the aforementioned problems, using the benchmark datasets as detailed in Section 4. Section 5 gives the different experimental details. Detailed ablation studies are discussed in Section 6, taking into account state-of-the-art techniques, while Section 7 concludes the paper.

2 Related works

Recent work by Dantcheva *et al.* [18] suggested that the accuracy of FR methods degrade by 76.21% due to make-up. It concludes that facial cosmetics can dramatically change facial appearance, both locally and globally, by altering colour, contrast, and texture. In fact, the impact due to the application of eye make-up is considered to be more pronounced than lipstick make-up [17–19]. The combination of eye and lipstick make-ups poses a greater challenge than individual ones. To date, there is limited scientific literature available for addressing the challenge of make-up induced changes. Hu *et al.* [20] used canonical correlation analysis (CCA) along with a support vector machine (SVM) classifier to facilitate the matching of face images before and after make-up. Guo *et al.* [21] learned the mapping between features extracted from patches in the before- and after-make-up images in order to minimise the disparity between them. The mapping was learned using CCA, regularised CCA, and partial least squares (PLS) methods. While mapping-based methods have been shown to be effective, they have two major limitations. First, the mapping between facial images before and after make-up can be complex,

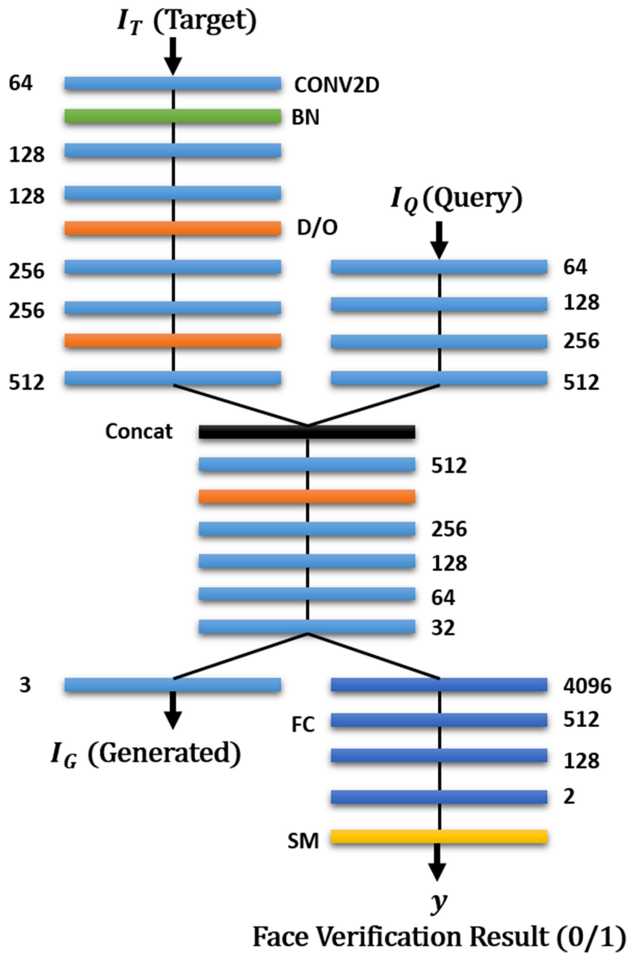

Fig. 1 *Proposed MakeUpMirror Architecture. CONV2D refers to the CONV2D layers. B/N – batch normalisation; D/O – dropout layers; FC – fully connected layers; SM – soft-max binary classification layer. Each of these layers follow a differentiable colour-coding (best viewed in colour).* I_T *and* I_Q *represent the faces with and without make-up, respectively, while* I_G *depicts the generated make-up face*

spatially non-invariant and non-linear. Therefore, it is insufficient to learn a single mapping in order to describe the complex relationship between before- and after-make-up samples [22]. Second, CCA and PLS methods often overfit the training data and fail to generalise well on unseen subjects [23]. A Gabor Fisher Classifier [24] method for FR, robust to illumination and facial expression variability, applies the enhanced Fisher linear discriminant model to an augmented Gabor feature vector derived from the Gabor wavelet representation of face images.

Unlike previous work [9, 25, 26], where commonly used make-up was observed to affect FR systems by obscuring a person's identity, we consider the scenario of generating faces with make-up used to alter the facial appearance. Chen *et al.* [27] introduced a patch-based ensemble learning method (alias, SRS-LDA), which uses multiple subspaces generated by sampling patches from before-make-up and after-make-up face images to address the problem of post make-up automated FR. Chen *et al.* [17] presented an automated make-up detection approach used to adaptively modify images prior to performing FR. Commercial Off-The-Shelf (COTS) Systems for FR usually referred to as COTS-1 [28], COTS-2 [28, 29], and COTS-3 [28, 30] are the leading commercial matchers [17, 31] providing state-of-the-art performance for FR. OpenBR [32] is a publicly available toolkit for biometric recognition and evaluation. The default FR algorithm in OpenBR is developed based on the Spectrally Sampled Structural Subspaces Features (4SF) [32]. Histogram of Monogenic Binary Pattern (HMBP) [33] is constructed by concatenating the histograms of monogenic binary pattern (MBP) from all subregions of a face. A novel texture descriptor, Histogram of Gabor Ordinal Measures (HGORM) [34], was developed to inherit the advantages from Gabor features and Ordinal Measures. HGORM applies Gabor filters of different orientations and scales on the face image and then computes Ordinal Measures over each Gabor magnitude response [35].

The works proposed in [36–38] mainly deal with multi-stage complex systems, using PCA (principal component analysis) for dimensionality reduction, followed by classification using SVM on the convolutional features obtained from their model. Zhu *et al.* [38] tries to warp faces into a canonical frontal view using a deep network for efficient classification. PCA on the network output in conjunction with an ensemble of SVMs is used for FV. Taigman *et al.* [37] proposed a siamese network to perform the task of FV. The compact network proposed by Sun *et al.* [36, 39, 40] uses an ensemble of 25 of these networks, each operating on a different face patch. The FaceNet proposed by Schroff *et al.* [41] uses a deep CNN to directly optimise the embedding itself, based on the triplet loss formulated by a triplet mining method. Duan *et al.* [42] proposed a new learning method for heterogeneous domain adaptation, in which the data from the source and target domains are represented by heterogeneous features with different dimensions. Using two different projection matrices, they first transform the data from two domains into a common subspace in order to measure the similarity. In our earlier work [10], an optimal feature-kernel combination had been used for adaptation to the target domain. Results were shown on three benchmark real-world surveillance face datasets.

All the fore-mentioned methods fail to verify faces under make-up, as they match the feature distributions (mostly used in DA) of the faces. They also fail to localise the cosmetic changes on a face and transfer that to query images. Many commercial make-up applications, e.g. TAAZ [43] etc., require human intervention to localise the exact boundaries of eyes and lips. Our proposed method automatically localises the facial attributes and transfers the cosmetic changes on the target, preserving the identity of the query image. A similar approach had also been proposed in [44], which automatically applies make-up on the masked facial parts based on rules, but they do not holistically transfer make-up from one face to another. Our proposed method is an end-to-end system, with no manual interference. Verification of faces with and/or without make-up is intrinsically done (matched) at the common pathway of the architecture (see Fig. 1), for make-up transfer and FV.

3 Proposed architecture for FV

The proposed architecture of SCNN is illustrated in Fig. 1. The inputs to the two streams in the network are denoted as I_T (the target face with make-up during training and both with or without make-up during testing) and I_Q (the query image without make-up during training and both with or without make-up during testing), where they correspond to samples from the same (or different) class(es). As evident from the illustration, the proposed architecture simultaneously deals with two problems at hand. The output at the left branch of the network generates a face image, I_G, with make-up corresponding to the identity of input probe (query) face (I_Q) without make-up. The right branch performs the regular task of FV between the two input images. The common stream at the middle of the network is instrumental to map the make-up information from I_T to I_Q.

The proposed architecture with two streams at the input consists of six 2D-convolutional (*CONV2D*) layers with increasing number of filters from 64 to 512, two dropout (*D*/*O*) layers with a dropout rate of 0.5 and one batch Normalisation layer (*BN*) at the left stream. The input right stream comprises of four CONV2D layers with the ascending number of filters from 64 to 512, similar to the left stream. The left is made deeper in order to capture the complex make-up elements that must be extracted from I_T and transferred to I_Q to generate I_G. The *BN* layers help in illumination normalisation of the faces present in the match, facilitating the increase the prominence of the cosmetics present in the exemplar face (I_T). As evident in the different CNN architectures proposed in the recent past [45], the *D*/*O* layers in our architecture prevent the overfitting of the model. Since, our model can map eye, lipstick and full make-up interchangeably over face images, dropout regularisation helps the network to avoid overfitting to a particular kind of make-up or dataset, thereby increasing the generalisability.

The outputs of the two streams are concatenated into a common pathway within the network. This fusion stream consists of five CONV2D layers with decreasing numbers of filters from 512 down to 32 and one dropout layer with a dropout rate of 0.25. These filters act as feature extractors or represent the feature maps present in the image. Deeper into the network, these filters capture complex higher-level features, to perform FV tasks. Higher numbers of parameters are involved in these layers with a larger number of filters, which may occasionally make the network unstable when trained using a relatively small-size dataset. The optimal number of filters necessary to effectively attenuate cosmetics parts of the face and transfer them to a target face appearance given for make-up are empirically found to be 32 in a single face image, thereby reducing the number of trainable parameters in the network. This common pathway acts as the backbone of the architecture which helps to transfer the make-up from the exemplar image to the test probe.

The output of common pathway is branched into two stems: the left stem outputs the probe with make-up, while the right stem verifies whether the two facial images at the input (I_T and I_Q) are of the same identity or not. The left stem has a CONV2D layer with the filter size of 3 which represents the three channels of an *RGB* image. The right stem acts as the regressor for the FV task, consisting of four fully connected convolutional layers and a soft-max regression (*SM*) layer for binary classification.

Our proposed architecture is motivated from the work in [46]. The left stream, which is a bit longer than the other, captures the cosmetic information present on the face along with the discriminative features of the face. The right stream captures only the discriminative information on the face. Both these information are then fused together in the common pathway, where the cosmetic information is transferred from I_T to I_Q to produce I_G, and the discriminative information required for FV is forwarded to the decision network in the right stem of the architecture (see Fig. 1), using the shared weights of the streams in the common pathway. The reconstruction loss, J_r (see (2)), is used to fine-tune the network for the purpose of transferring the make-up in the target image to the query image in the left stem of the model. The decision network at the right stem uses the shared weights from the common pathway to produce the verification result, y, which is also fine-tuned using the *softmax* loss, $\mathscr{L}$, in the decision network in the right stem. To summarise, the two streams of the *MakeUpMirror* help to capture the discriminative parts of the face, where the fusion stream transfers the make-up from the target to the query face, to be provided at the left stem, while only the verification is done at the right stem.

3.1 Analytical details of SCNN

The proposed SCNN architecture is developed in *Keras* with *TensorFlow* backend. The training process involves the minimisation of the following objective function ($\mathscr{J}$), using back-propagation, given as follows:

$$\mathscr{J} = J_r\left(I^{(\mathrm{gen})}, I^{(\mathrm{gt})}\right) + \alpha \mathscr{L}\left(\theta, \xi^{(v)}\right) \tag{1}$$

The first term in the objective function is the reconstruction error between the generated image sample (I^{gen}) with make-up when compared to its ground-truth (I^{gt}) as follows:

$$J_r\left(I^{(\mathrm{gen})}, I^{(\mathrm{gt})}\right) = \sum_{i=1}^{n_T} \| I_i^{(\mathrm{gen})} - I_i^{(\mathrm{gt})} \|^2 \tag{2}$$

where n_T represents the total number of training samples. In our case, $I^{(\mathrm{gen})} \equiv I_G$ and $I^{(\mathrm{gt})}$ correspond to the make-up image from the target set, with the same identity as I_Q. The second term in $\mathscr{J}$ is the loss function of soft-max regression to perform the binary classification for FV. Specifically,

$$\mathscr{L}\left(\theta, \xi^{(v)}\right) = -\frac{1}{n_T} \sum_{i=1}^{n_T} \sum_{j=0}^{1} 1\{y_i^{(v)} = j\} \log \frac{e^{\theta_j^T \xi_i^{(v)}}}{\sum_{l=0}^{1} e^{\theta_l^T \xi_i^{(v)}}} \tag{3}$$

where, $\xi_i^{(v)}$ is the output of the two-node layer preceding the SM; $\theta_j^T (j \in \{0, 1\})$ is the jth row of $\boldsymbol{W}$ and the verification output $y_i^{(v)}$ is given by

$$y_i^{(v)} = f\left(\boldsymbol{W} \xi_i^{(v)} + \boldsymbol{b}\right) \tag{4}$$

where f is a non-linear activation function (ReLU [47] in this case), $\boldsymbol{W} \in \mathbb{R}^{2 \times 1}$ is the weight matrix while $\boldsymbol{b} \in \mathbb{R}^{2 \times 1}$ is a bias vector. The value α (set as 0.5, empirically found to provide the best performance) is the weight to highlight the relative significance between the pair of terms in (1). The model is trained for 70K epochs, with a batch-size of 20 face samples. The overall proposed architecture achieves considerable improvement in accuracy, measured using *equal error rates* (EERs), when experimented using three real-world make-up datasets.

4 Datasets used for experimentation

Experiments have been carried out over three benchmark real-world make-up face datasets and one real-world surveillance dataset (FR_SURV) with high-resolution (HR) gallery and low-resolution (LR) probe images. Each of the three benchmark make-up datasets consists of tightly cropped face images with and without make-up, details of which are summarised in Table 1. The real-world surveillance face dataset has only been used for the make-up transfer to check the effectiveness of our proposed model when the ethnicity of the person changes. The different protocols (refer Section 5) used for experimentations, using these datasets, are adapted from [31].

Table 1 Dataset description

Dataset	# Subjects	Images/subject	# Images
YMU [17, 18]	151	4 (2 without and 2 with make-up)	604
VMU [18]	51	5 (2 without make-up, 1 with lipstick, 1 with eye make-up and 1 with full make-up)	255
MIW [17]	125	1 (or) 2	154 images (77 with make-up and 77 without make-up)
FR_SURV [48]	51	40 (20 HR and 20 LR)	1020

5 Experimental details

Experiments on the aforementioned three datasets are performed using the three protocols as given in [17, 27, 31], briefly described below.

5.1 Protocol 1 (baseline for YMU)

The YMU dataset consists of four face images for each subject: two (N_1 and N_2) belonging to those without make-up and the remaining two (M_1 and M_2) with make-up. Genuine and impostor pairs are formed for three cases using image pairs according to the following:

1. Matching N_1 against N_2 (N versus N): both image samples do not have make-up.
2. Matching M_1 against M_2 (M versus M): both image samples have make-up.

3. Matching of four combinations as: M_i against N_j, $i, j \in \{1, 2\}$ (N versus M): one with make-up (M_i) while the other has no make-up (N_j).

A total of 906 genuine and 181,200 impostor image pairs are formed for the three cases mentioned above, as follows:

1. 151 genuine and 45,300 impostor image pairs,
2. 151 genuine and 45,300 impostor image pairs, and
3. 604 genuine and 90,600 impostor image pairs.

5.2 Protocol 2 (baseline for VMU)

The VMU dataset comprises of five images per subject: two original images without make-up (denoted as N) and three images after adding make-up on one of the original images using the publicly available tool from *TAAZ* [43], viz. (i) application of lipstick only (denoted as L); (ii) application of eye make-up only (denoted as E); and (iii) application of a full make-up consisting of lipstick, foundation, blush and eye make-up (denoted as F). Three variants of experimentation have been performed for FV on this dataset, where an image (N) without make-up is compared against an image with varied make-ups (of the same/different individual/subject) as listed below:

1. N versus L: lip make-up.
2. N versus E: eye make-up.
3. N versus F: full make-up.

For each of the three cases above, 102 genuine and 5100 impostor image pairs were used for experimentation.

5.3 Protocol 3 (chimeric evaluation)

In this evaluation protocol, the YMU and VMU datasets have been combined (concatenated) together for training, resulting in 910 genuine and 182,910 impostor image pairs. The SCNN is trained using pairs of images with and without make-up, to replicate the make-up (I_T) without altering the identity of the subject in I_Q (see Fig. 1). In this mode of evaluation, experiments have been performed to test the generalisation of our proposed architecture in mirroring make-ups.

The ratio of the number of training versus testing samples is maintained at 70 : 30 throughout all experimentations. The training, as proposed in Chen *et al.* [17, 27, 31], uses both genuine ($y = 1$), and impostor ($y = 0$) image pairs. Training of the *MakeUpMirror* architecture, as given in Fig. 1, is done using a set of quadruplets, $\{I_Q, I_T, I_G, y\}$. Here, I_G represents the ground-truth face image with make-up of the same identity as the query image, I_Q (without make-up). y gives the verification label (0/1) and I_T is the target face with make-up. From these datasets, which are organised as two partitioned sets of impostor and genuine pairs, 70% of these pairs from each sets are randomly selected to form the quadruplets for training with the various protocols discussed above, and the remaining 30% of the samples are used for testing. The training and testing sets never overlap. During testing, both I_Q and I_T may appear with or without make-up on faces, depending on the protocol being used for verification.

6 Performance evaluation

The performance of the proposed model measured using the EER is compared with several benchmark or state-of-the-art techniques as evaluated and reported in [31] (lower the better). Several commercial and academic algorithms, viz. COTS-1, COTS-2, COTS-3, OpenBR [32], Local Gabor Binary Pattern (LGBP) [35], Local Gradient Gabor Pattern (LGGP) [19], SRS-LDA [27] and HMBP [33], have been used for comparison with our proposed model. Fig. 2*a* shows the Receiver-Operator Characteristics (ROC) curves for our proposed architecture on the three cases of *Protocol 1*. The EER values are obtained at the intersections of the black dotted line with the corresponding curves. For comparison of performance of our method with the recent state-of-the-art methods, *case 3 of Protocol 1* is considered. The red ROC curve at the bottom of Fig. 2*b* shows a lower EER value for our method, compared to the other methods. Thus, our proposed method outperforms all other methods. Table 2 gives the EER of the model trained on YMU based on the *Protocol 1* (see Section 5.1). The results marked in bold reveal that our proposed method outperforms all the other competing methods. The results depicted in the last column are the toughest testing scenario. The task of FV between a make-up face and that without make-up is more complex to handle even when both the images are from the same category, as their features come from different distributions. The complexity increases since the N versus M scenario falls under the problem of cross-domain image matching. The cosmetic changes on the faces make it harder for the supervised algorithms to perform the task of FV efficiently between a face before and after make-up. Our SCNN model captures the identity of the face better than the other methods being compared, as shown in Table 2. The second best method is SRS-LDA [27], which is based on the multi-patch-based ensemble learning method.

(a) *Protocol 1* (all three cases).
(b) *Protocol 1 – case 3* compared to the state-of-the-art methods.
(c) *Protocol 1 – case 3* compared to the five nearest competing state-of-the-art methods, showing genuine acceptance rate (GAR) at lower values of false acceptance rate (FAR).

Users of biometric authentication systems demand high performance at low values of FAR [49]. The performance of our proposed system, given by the red curve in Fig. 2*c*, shows a significant improvement in the performance in terms of GARs at very low values of FAR. Table 3 gives the performances for several state-of-the-art techniques discussed above and their combinations, compared with our proposed model, at a low value of FAR. For the fusion [27] of features, we have normalised the individual match scores generated by different matchers based on the min–max rule, followed by the simple sum rule. The table reports values for GAR at low FAR (0.1%) (shown by the black dotted line in Fig. 2*c*). The third column reports the EER values (for only N versus M, *Protocol 1 – case 3*) of the methods, which partly overlap with *column 4* of Table 2. The proposed architecture outperforms all other methods and their combinations by a considerable margin, as evident from the results shown in bold in the last row. Higher GAR values at a low FAR are considered better for benchmarking a biometric authentication system. In the eighth row of Table 3, a method obtained by fusing (at score-level) SR-LDA [27] and COTS-2 + COTS-3 algorithms performs the second best, which is based on hand-crafted features.

Fig. 3 shows the ROC curves for our proposed architecture on the three cases of *Protocol 2*. The EER values are values are calculated at the intersections of the black dotted line with the curves. Table 4 gives the EER values of the models trained on the VMU datasets based on *Protocol 2*. The EER values of our proposed architecture are compared with several algorithms: COTS-1, COTS-3, LBP [2], Gabor [24], HMBP [33], LGGP [19], and LGBP [35], on the VMU dataset. The results marked in bold show that our proposed method performs the best. Again, the higher values of EER at the last column depicts that N versus F is the toughest scenario and the most complex case. As the make-ups in the VMU dataset are generated synthetically by a commercial software *TAAZ*, it is uniform and highly predictable. Owing to this reason and the smaller size of the dataset, SCNN provides better performance on VMU than YMU dataset. The COTS-based methods perform the second best for this dataset, which also proves the unbiased nature of our model to a dataset.

6.1 Qualitative comparison of make-up transfer on face

As the models are trained based on the three protocols discussed in Section 5, it is observed that not only the results of FV vary, but also the make-up generation on the non-make-up faces varies accordingly. Qualitative results are depicted in Figs. 4–6 using four sub-figures (row-wise) each with three images, viz. the make-up

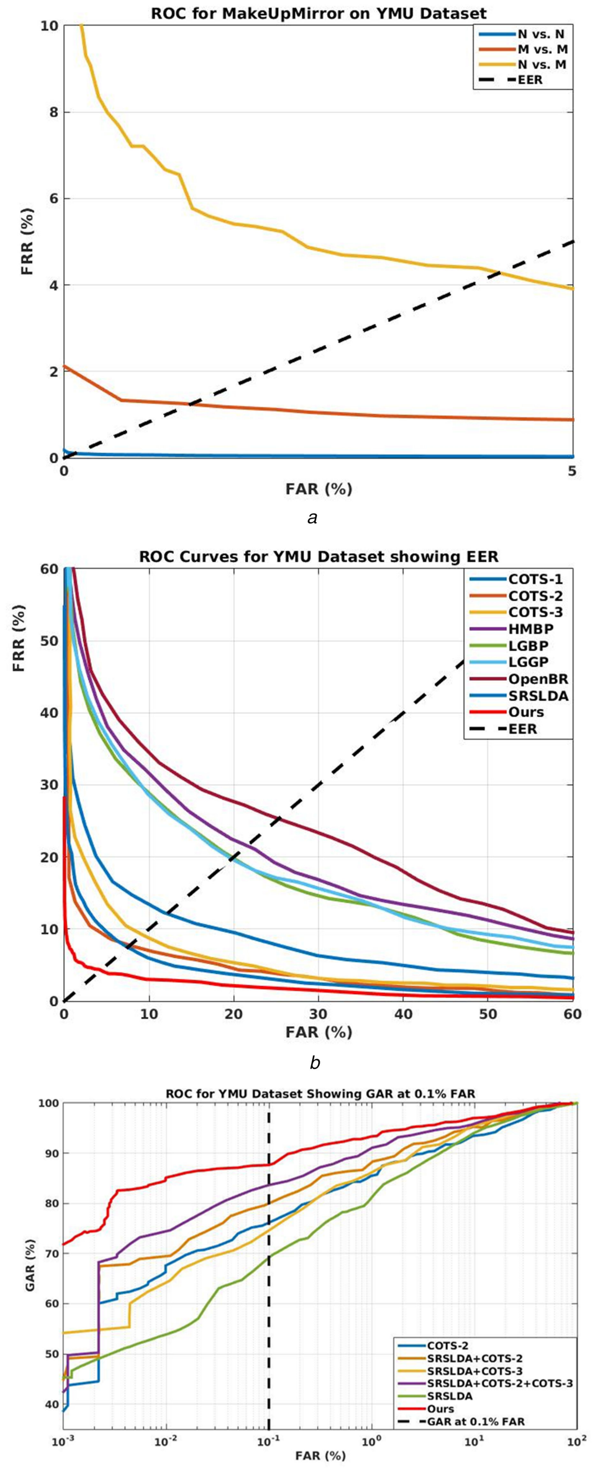

Fig. 2 *ROC curves for the YMU dataset. (All plots in (b) & (c) except our proposed method in 'red' are directly obtained from [27, 31])*

image (I_T), face with no make-up (I_Q) and generated make-up (I_G) obtained at the output layer of our proposed model given in Fig. 1. The four sub-figures depict the results of each protocol on four different datasets.

Fig. 4 shows the results of training our architecture using *Protocol 1*. In this scenario, the proposed architecture is trained using all the genuine and impostor images from the YMU dataset. The model not only learns the cosmetic alterations on the make-up faces, but also a mapping of the different cosmetics from the make-up faces (I_T) to that without make-up (I_Q), to produce a full make-up face (I_G) that is perceived to be more attractive to the human eyes. Owing to the varied types of make-up on the YMU face dataset, the model generates a make-up face based on the skin-tone

Table 2 Equal error rates (%) on the YMU dataset [17, 18], using *Protocol 1*

	N versus *N*	*M* versus *M*	*N* versus *M*
COTS-1	3.845	7.079	12.04
COTS-2	0.6912	1.327	7.692
COTS-3	0.1141	3.291	9.178
OpenBR [32]	6.872	16.44	25.2
LGBP [35]	5.345	8.773	19.71
LGGP [19]	5.359	8.007	19.7
HMBP [33]	6.25	10.87	21.54
SRS-LDA [27]	0.62	1.99	7.59
proposed (SCNN)	**0.09**	**1.24**	**4.28**

The values of all the rows except the last row are obtained from [27, 31].

Table 3 GAR performance with FAR at 0.1% and equal error rates (%) on the YMU dataset [17, 18], for different methods (*Protocol 1–case 3*) and fusion of a few of them, along with the proposed method

Algorithms	GAR (%)	EER (%)
COTS-1	48.86	12.04
COTS-2	76.15	7.69
COTS-3	58.48	9.18
SRS-LDA [27]	69.24	7.59
SRS-LDA + COTS-2	79.88	5.98
SRS-LDA + COTS-3	74.44	6.59
COTS-2 + COTS-3	80.54	5.98
SRS-LDA + COTS-2 + COTS-3	83.61	5.19
LGGP [19]	32.82	20.48
HGORM [34]	37.99	20.24
DS-LBP [27]	30.26	19.65
SRS-LDA + LGGP	57.99	11.01
SRS-LDA + HGORM	63.64	10.21
SRS-LDA + DS-LBP	58.06	11.35
proposed (SCNN)	**87.65**	**4.28**

The values of all the rows except the last row are obtained from [27].

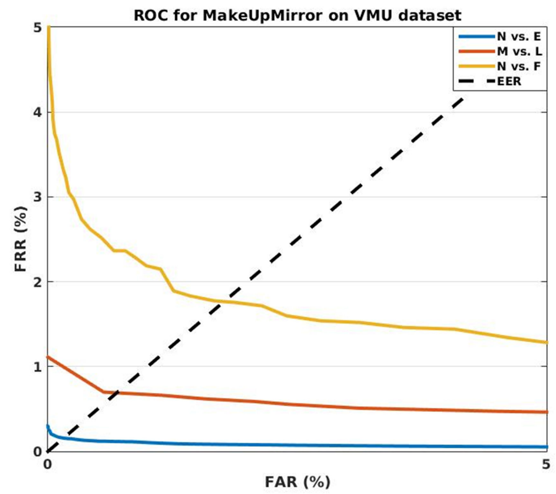

Fig. 3 *ROC curve for the VMU dataset showing the performance of our method for Protocol 2 (all three cases)*

and facial layout and structures. The trained model is able to capture the cosmetic changes in the face that are present in the YMU dataset. The original (target) make-up face image I_T (see Fig. 1) is taken from the YMU dataset, while a sample each from all the datasets as given in Table 1 are without make-up I_Q (see Fig. 1). The notation (**A-B**) below each sub-figure denotes the input face datasets; as **A** is the dataset from which the original image, I_T, with make-up (YMU in this case) is taken and **B** denotes the dataset

Table 4 Equal error rates (%) on the VMU dataset [18], using Protocol 2

	N versus *L*	*N* versus *E*	*N* versus *F*
COTS-1	1.97	0.9849	1.959
COTS-3	0.4805	1.984	2.248
LBP [2]	3.43	4.32	4.79
Gabor [24]	8.56	8.46	11.38
HMBP [33]	5.0	10.87	8.05
LGGP [19]	4.805	8.688	8.164
LGBP [35]	2.9	5.44	5.42
proposed (SCNN)	**0.15**	**0.68**	**1.76**

The values of all the rows except the last row are obtained from [27, 31].

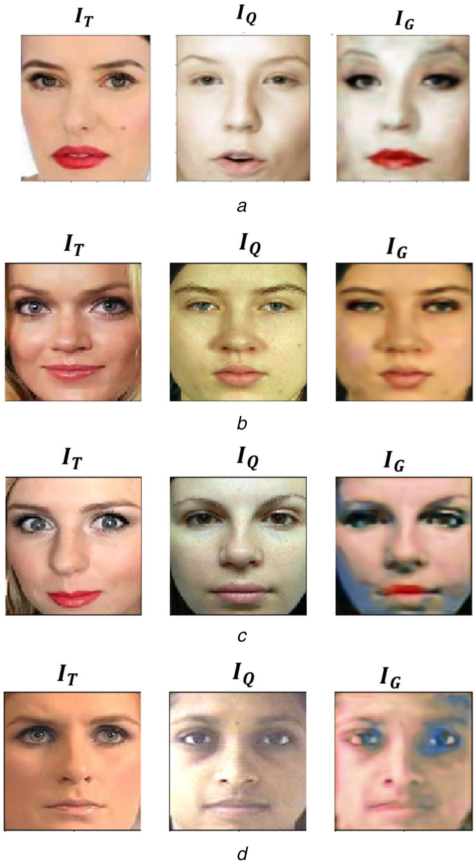

Fig. 4 *Images showing the results of make-up transfer using the model trained on the YMU dataset*
(a) YMU–YMU, ***(b)*** YMU–VMU, ***(c)*** YMU–MIW, ***(d)*** YMU–EXT

from which the face image, I_Q, without make-up is considered. The corresponding generated image, I_G (see Fig. 1), obtained from these datasets is used to observe the quality in transfer of make-up. The dataset denoted as '*EXT'* at the last row of Fig. 4 refers to an image from the gallery samples of FR_SURV [48] dataset. The first row in Fig. 4, where the target face, I_Q, without make-up is chosen from the YMU dataset, exhibits the best mirroring of the make-up. However, to determine the effectiveness of our architecture across datasets, results on different target facial images from other datasets are also shown. The results across datasets qualitatively exhibit the scope of applicability of our proposed method with varying environmental conditions and ethnicity.

Similar results for *Protocol 2* are shown in Fig. 5 for samples trained on the VMU [18] dataset and arranged as described above for the case of *Protocol 1* (i.e. as shown in Fig. 4). The model here captures the synthetically generated cosmetic alterations. The make-up on these face samples are mostly uniform, since they are synthetically (manually) created using the commercial application *TAAZ* [43]. The model generated better results for the VMU dataset in Fig. 5 than the natural make-up images of the YMU dataset in Fig. 4. The efficiency in transferability of the make-up on face parts as: lips and the eyes are more vivid in Fig. 5, since, in this case, the VMU dataset [18] has a subset of make-up which is localised only on the lips or eyes.

Protocol 3 trains our proposed SCNN on a chimeric dataset formed by combining YMU and VMU datasets. This protocol trains the model with faces from two different environmental settings under which the faces are captured. Ablation studies show that SCNN takes care of the localisation of cosmetic alterations of the face sample, as well as preserves the contextual information. The model can learn the cosmetic changes of both the datasets simultaneously, and their results are displayed in Fig. 6, using a similar layout as in Figs. 4 and 5.

Next, our proposed model was subjected to rigorous testing conditions where the make-up is generated on the degraded or low-resolution images, using *Protocol 3*. Our proposed architecture generates cosmetic changes in the low-resolution faces with considerable success, and the qualitative results are shown in Fig. 7. Here, *SURV* denotes the FR_SURV dataset [48] from which the probe images (with low resolution) are obtained. These are captured in an unconstrained outdoor environment using surveillance cameras. The FR_SURV dataset consists of faces of subjects with different ethnicity than YMU and VMU. The cross-ethnic comparison highlights the effectiveness of the model to transfer facial make-up irrespective of the variations of the inherent characteristics on the face. Results in Fig. 7 show that our proposed model was also able to produce encouraging results on low-resolution images (probes).

Fig. 8 shows patch-wise correspondence of the pair of images I_T and I_G (see Fig. 1). For comparison, here we have taken I_T and I_G belonging to the same subject. The images represent three parts of the face, where the effect of cosmetics on the faces is most evident. The target images for comparison, which are only available in the test-sets of the VMU and YMU datasets. Fig. 8 shows that the results (cosmetic transfer of make-up) of our proposed methods are comparable to that synthetically generated by a commercial software, like *TAAZ* [43]. Note that the target images in the VMU dataset are generated, using *TAAZ*, using manual localisation of facial attributes.

To evaluate this quantitatively, we have used the peak signal-to-noise ratio (PSNR) [50] and structural similarity (SSIM) [51] metrics on the images of YMU and VMU datasets, on which make-up is applied. Table 5 gives two quantitative measures to measure the quality of the images generated by the model with respect to its ground-truth. First among them is the PSNR [50], which captures the difference in the pixel values between the two images. Secondly, SSIM [51] index was also used to quantify the generated results, which estimates the holistic similarity between two images. The generated images I_G (i.e. I in (5) and (6)) are compared with the corresponding ground-truths I_T (i.e. K in (5) and (6)), available in the YMU and VMU datasets (the other two datasets do not have make-up ground-truth data). PSNR (higher the better) is defined as follows:

$$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}(I)^2}{\mathrm{MSE}(I, K)}\right) \tag{5}$$

where $\mathrm{MSE}(I, K) = (1/mn)\sum_{i=1}^{m-1}\sum_{i=1}^{n-1}[I(i, j) - K(i, j)]^2$, I and K are of the size $m \times n$ pixels. The SSIM index [51] is calculated on multiple images, which is briefly described below. The measure between the two images I and K, of common size $m \times n$ pixels, is:

$$\mathrm{SSIM}(I, K) = \frac{(2\mu_I\mu_K + c_1)(2\sigma_{IK} + c_2)}{(\mu_I^2 + \mu_K^2 + c_1)(\sigma_I^2 + \sigma_K^2 + c_2)} \tag{6}$$

with μ_I, μ_K the average of I and K, respectively, σ_I^2, σ_K^2 their respective variances, σ_{IK} the covariance of I and K; $c_1 = (k_1L)^2$, $c_2 = (k_2L)^2$ are two variables used to stabilise the weak denominator, L is the dynamic range of the pixel values (typically this is $2^{\#\text{bits/pixel}} - 1$), whereas $k_1 = 0.01$ and $k_2 = 0.03$ are set by default. Results in Table 5 show that SCNN outperforms the

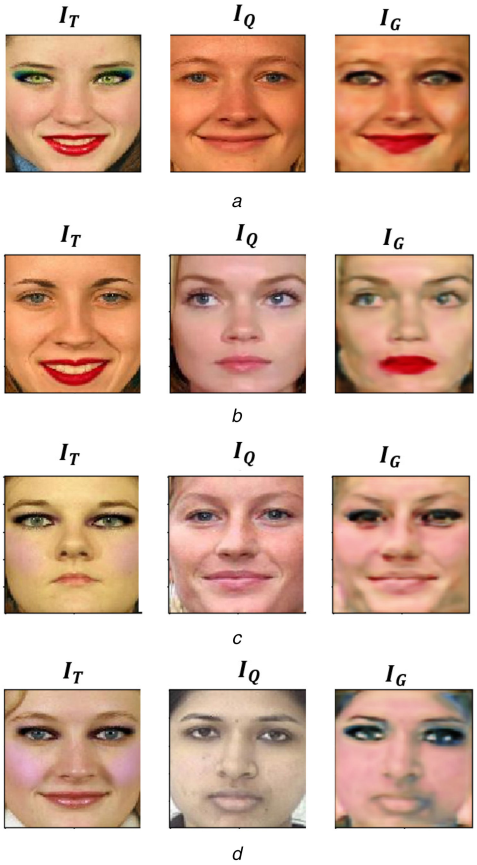

Fig. 5 *Images showing the results of make-up transfer using the model trained on the VMU dataset*
(a) VMU–VMU, ***(b)*** VMU–YMU, ***(c)*** VMU–MIW, ***(d)*** VMU–EXT

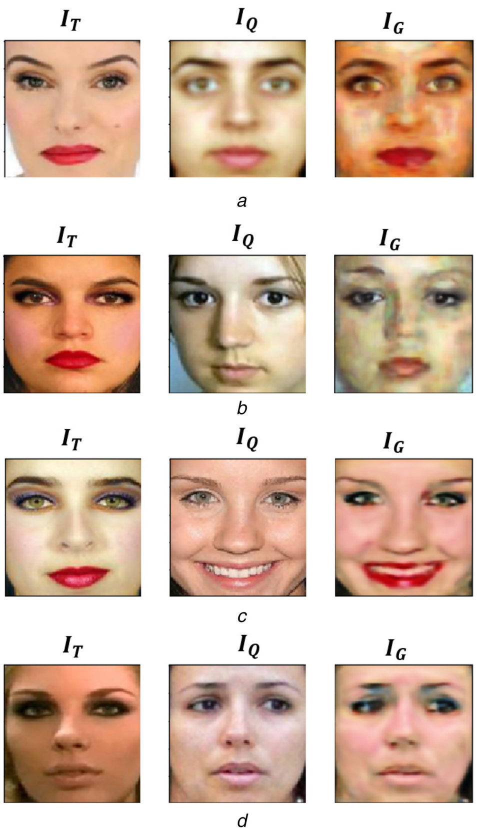

Fig. 6 *Images showing the results of make-up transfer using the model trained on chimeric dataset (Protocol 3)*
(a) YMU–VMU, ***(b)*** VMU–YMU, ***(c)*** VMU–MIW, ***(d)*** YMU–MIW

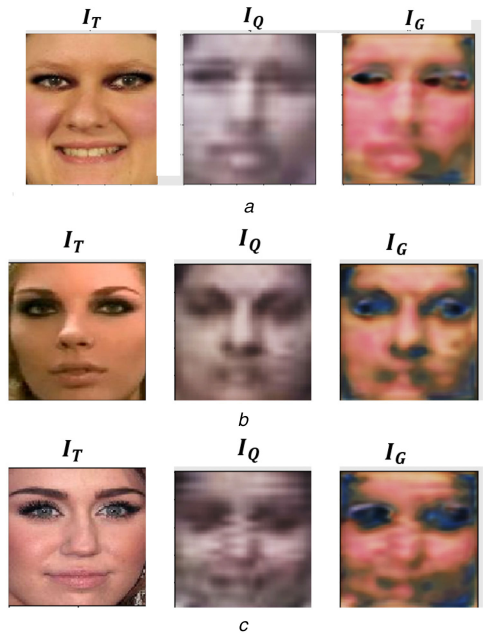

Fig. 7 *Images showing the results of make-up transfer on low-resolution surveillance images, from FR_SURV [48] dataset*
(a) VMU–SURV, ***(b)*** YMU–SURV, ***(c)*** MIW–SURV

Fig. 8 *Images showing the results of make-up transfer on discriminative patches of the face, from the VMU dataset*
(a) Original make-up image (from the VMU dataset, using *TAAZ*), ***(b)*** generated make-up image (I_G), generated by SCNN (proposed)

Table 5 Average PSNR/SSIM values on the datasets, comparing our proposed technique with the method proposed by Xu *et al.* [44]

Dataset	Proposed (SCNN)	Xu *et al.* [44]
VMU	16.47/0.57	13.26/0.43
YMU	14.57/0.66	12.29/0.51

method proposed by Xu *et al.* [44]. The superior performance of our proposed 'MakepUpMirror' model, shown in Tables 2–5, is due to the use efficient minimisation (1) function while training and architectural fusion of features from both pair of faces under make-up.

7 Conclusion

The proposed architecture not only outperforms other state-of-the-art benchmark techniques used for FV, but also automates the process of make-up transfer on a non-make-up face with considerable success. The proposed model identifies a person with make-up in real-world datasets, even in unconstrained scenarios. The SCNN provides a considerable improvement in GAR (4% at 0.1% FAR) and EER (∼10–42%) values in all the cases, even for low FAR values. The transfer of cosmetic variations to a face from that of another individual has been performed with no manual intervention. The transfer of make-up takes care of the face structure holistically, where the PSNR and SSIM values improve by ∼20–24 and ∼29–32%, respectively, when compared to a recent state-of-the-art technique. Several unconstrained faces under varied environmental conditions were used to verify the effectiveness of our method for FV application, with no human intervention. The proposed method can be further extended to deal with make-up removal tasks, overcome spoofing of faces due to make-up as well as for the identification of faces under make-up. Exploration of redesigning the model to work for fusing information in case of multi-modal biometric scenarios could be a challenging work for the researchers in the future.

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